

Namespace SixLabors.ImageSharp.Web.AWS

Classes

AmazonS3BucketClient

Represents a scoped Amazon S3 client instance that is explicitly associated with a single S3 bucket. This wrapper provides a strongly-typed link between the client and the bucket it operates on, and optionally manages the lifetime of the underlying Amazon.S3.AmazonS3 Client.

Interfaces

IAWSS3BucketClientOptions

Provides a common interface for AWS S3 Bucket Client Options.

Class AmazonS3BucketClient

Namespace: [SixLabors.ImageSharp.Web.AWS](#)

Assembly: SixLabors.ImageSharp.Web.Providers.AWS.dll

Represents a scoped Amazon S3 client instance that is explicitly associated with a single S3 bucket. This wrapper provides a strongly-typed link between the client and the bucket it operates on, and optionally manages the lifetime of the underlying [Amazon.S3.AmazonS3Client](#).

```
public sealed class AmazonS3BucketClient : IDisposable
```

Inheritance

[object](#)  ← [AmazonS3BucketClient](#)

Implements

[IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

AmazonS3BucketClient(string, AmazonS3Client, bool)

Initializes a new instance of the [AmazonS3BucketClient](#) class.

```
public AmazonS3BucketClient(string bucketName, AmazonS3Client client, bool disposeClient  
= true)
```

Parameters

[bucketName](#) [string](#) 

The bucket name associated with this client instance.

[client](#) [AmazonS3Client](#)

The underlying Amazon S3 client instance. This should be an already configured instance of Amazon.S3.AmazonS3Client.

`disposeClient` `bool`[↗](#)

A value indicating whether the underlying client should be disposed when this instance is disposed.

Properties

BucketName

Gets the bucket name associated with this client instance.

```
public string BucketName { get; }
```

Property Value

`string`[↗](#)

Client

Gets the underlying Amazon S3 client instance.

```
public AmazonS3Client Client { get; }
```

Property Value

AmazonS3Client

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

Interface IAWSS3BucketClientOptions

Namespace: `SixLabors.ImageSharp.Web.AWS`

Assembly: `SixLabors.ImageSharp.Web.Providers.AWS.dll`

Provides a common interface for AWS S3 Bucket Client Options.

```
public interface IAWSS3BucketClientOptions
```

Properties

AccessKey

Gets or sets the AWS key - Can be used to override keys provided by the environment. If deploying inside an EC2 instance AWS keys will already be available via environment variables and don't need to be specified. Follow AWS best security practices on <https://docs.aws.amazon.com/general/latest/gr/aws-access-keys-best-practices.html>.

```
string? AccessKey { get; set; }
```

Property Value

`string`

AccessSecret

Gets or sets the AWS endpoint - used to override the default service endpoint. If deploying inside an EC2 instance AWS keys will already be available via environment variables and don't need to be specified. Follow AWS best security practices on <https://docs.aws.amazon.com/general/latest/gr/aws-access-keys-best-practices.html>.

```
string? AccessSecret { get; set; }
```

Property Value

string[↗]

BucketName

Gets or sets the AWS bucket name. Cannot be **null**[↗] when **S3ClientFactory** is not set.

```
string BucketName { get; set; }
```

Property Value

string[↗]

Endpoint

Gets or sets the AWS endpoint - used for testing to over region endpoint allowing it to be set to localhost.

```
string? Endpoint { get; set; }
```

Property Value

string[↗]

Region

Gets or sets the AWS region endpoint (us-east-1/us-west-1/ap-southeast-2).

```
string? Region { get; set; }
```

Property Value

string[↗]

S3ClientFactory

Gets or sets a custom factory method to create an **AmazonS3BucketClient**.

```
Func<IAWSS3BucketClientOptions, IServiceProvider, AmazonS3BucketClient>? S3ClientFactory {  
    get; set; }
```

Property Value

Func[↗](#)<IAWSS3BucketClientOptions, IServiceProvider[↗](#), AmazonS3BucketClient>

Timeout

Gets or sets a value indicating the timeout for the S3 client. If the value is set, the value is assigned to the Timeout property of the HttpRequest/HttpClient object used to send requests.

```
TimeSpan? Timeout { get; set; }
```

Property Value

TimeSpan[↗](#)?

UseAccelerateEndpoint

Gets or sets a value indicating whether the S3 accelerate endpoint is used. The feature must be enabled on the bucket. Follow AWS instruction on

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/transfer-acceleration.html>[↗](#).

```
bool UseAccelerateEndpoint { get; set; }
```

Property Value

bool[↗](#)

Namespace SixLabors.ImageSharp.Web. AWS.Caching

Classes

`AWSS3StorageCache`

Implements an AWS S3 Storage based cache.

`AWSS3StorageCacheOptions`

Configuration options for the `AWSS3StorageCache` provider.

Class AWSS3StorageCache

Namespace: [SixLabors.ImageSharp.Web.AWS.Caching](#)

Assembly: [SixLabors.ImageSharp.Web.Providers.AWS.dll](#)

Implements an AWS S3 Storage based cache.

```
public class AWSS3StorageCache : IImageCache, IDisposable
```

Inheritance

[object](#)  ← [AWSS3StorageCache](#)

Implements

[IImageCache](#), [IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

AWSS3StorageCache(IOptions<AWSS3StorageCacheOptions>, IServiceProvider)

Initializes a new instance of the [AWSS3StorageCache](#) class.

```
public AWSS3StorageCache(IOptions<AWSS3StorageCacheOptions> cacheOptions,  
    IServiceProvider serviceProvider)
```

Parameters

cacheOptions [IOptions](#)  <[AWSS3StorageCacheOptions](#)>

The cache options.

serviceProvider [IServiceProvider](#) 

The current service provider.

Methods

CreateIfNotExists(AWSS3StorageCacheOptions, S3CannedACL)

Creates a new bucket under the specified account if a bucket with the same name does not already exist.

```
public static PutBucketResponse? CreateIfNotExists(AWSS3StorageCacheOptions options,  
S3CannedACL acl)
```

Parameters

options *AWSS3StorageCacheOptions*

The AWS S3 Storage cache options.

acl *S3CannedACL*

Specifies whether data in the bucket may be accessed publicly and the level of access. Amazon.S3.S3CannedACL.PublicRead specifies full public read access for bucket and object data. Amazon.S3.S3CannedACL.Private specifies that the bucket data is private to the account owner.

Returns

PutBucketResponse

If the bucket does not already exist, a Amazon.S3.Model.PutBucketResponse describing the newly created bucket. If the container already exists, **null** .

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

Dispose(bool)

Releases the unmanaged resources used by the `AWSS3StorageCache` and optionally releases the managed resources.

```
protected virtual void Dispose(bool disposing)
```

Parameters

`disposing bool` [↗](#)

true to release both managed and unmanaged resources; false to release only unmanaged resources.

GetAsync(string)

Gets the image resolver associated with the specified key.

```
public Task<IImageCacheResolver?> GetAsync(string key)
```

Parameters

`key string` [↗](#)

The cache key.

Returns

`Task` [↗](#) `<IImageCacheResolver>`

The `IImageResolver`.

SetAsync(string, Stream, ImageCacheMetadata)

Sets the value associated with the specified key.

```
public Task SetAsync(string key, Stream stream, ImageCacheMetadata metadata)
```

Parameters

key [string](#)

The cache key.

stream [Stream](#)

The stream containing the image to store.

metadata [ImageCacheMetadata](#)

The [ImageCacheMetadata](#) associated with the image to store.

Returns

[Task](#)

The task.

Class AWSS3StorageCacheOptions

Namespace: [SixLabors.ImageSharp.Web.AWS.Caching](#)

Assembly: [SixLabors.ImageSharp.Web.Providers.AWS.dll](#)

Configuration options for the [AWSS3StorageCache](#) provider.

```
public class AWSS3StorageCacheOptions : IAWSS3BucketClientOptions
```

Inheritance

[object](#)  ← [AWSS3StorageCacheOptions](#)

Implements

[IAWSS3BucketClientOptions](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Properties

AccessKey

Gets or sets the AWS key - Can be used to override keys provided by the environment. If deploying inside an EC2 instance AWS keys will already be available via environment variables and don't need to be specified. Follow AWS best security practices on

<https://docs.aws.amazon.com/general/latest/gr/aws-access-keys-best-practices.html> .

```
public string? AccessKey { get; set; }
```

Property Value

[string](#) 

AccessSecret

Gets or sets the AWS endpoint - used to override the default service endpoint. If deploying inside an EC2 instance AWS keys will already be available via environment variables and don't need to be specified. Follow AWS best security practices on

<https://docs.aws.amazon.com/general/latest/gr/aws-access-keys-best-practices.html>.

```
public string? AccessSecret { get; set; }
```

Property Value

string

BucketName

Gets or sets the AWS bucket name. Cannot be `null` when `S3ClientFactory` is not set.

```
public string BucketName { get; set; }
```

Property Value

string

CacheFolder

Gets or sets the cache folder's name that'll store cache files under the configured bucket.

```
public string? CacheFolder { get; set; }
```

Property Value

string

Endpoint

Gets or sets the AWS endpoint - used for testing to over region endpoint allowing it to be set to localhost.

```
public string? Endpoint { get; set; }
```

Property Value

string [↗](#)

Region

Gets or sets the AWS region endpoint (us-east-1/us-west-1/ap-southeast-2).

```
public string? Region { get; set; }
```

Property Value

string [↗](#)

S3ClientFactory

Gets or sets a custom factory method to create an [AmazonS3BucketClient](#).

```
public Func<IAWSS3BucketClientOptions, IServiceProvider, AmazonS3BucketClient>?  
S3ClientFactory { get; set; }
```

Property Value

Func [↗](#)<IAWSS3BucketClientOptions, IServiceProvider [↗](#), AmazonS3BucketClient>

Timeout

Gets or sets a value indicating the timeout for the S3 client. If the value is set, the value is assigned to the Timeout property of the HttpWebRequest/HttpClient object used to send requests.

```
public TimeSpan? Timeout { get; set; }
```

Property Value

TimeSpan[↗](#)?

UseAccelerateEndpoint

Gets or sets a value indicating whether the S3 accelerate endpoint is used. The feature must be enabled on the bucket. Follow AWS instruction on

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/transfer-acceleration.html>[↗](#).

```
public bool UseAccelerateEndpoint { get; set; }
```

Property Value

bool[↗](#)

Namespace SixLabors.ImageSharp.Web. AWS.Providers

Classes

AWSS3BucketClientOptions

Configuration options for the [AWSS3StorageImageProvider](#) provider.

AWSS3StorageImageProvider

Returns images stored in AWS S3.

AWSS3StorageImageProviderOptions

Configuration options for the [AWSS3StorageImageProviderOptions](#) provider.

Class AWSS3BucketClientOptions

Namespace: [SixLabors.ImageSharp.Web.AWS.Providers](#)

Assembly: [SixLabors.ImageSharp.Web.Providers.AWS.dll](#)

Configuration options for the [AWSS3StorageImageProvider](#) provider.

```
public class AWSS3BucketClientOptions : IAWSS3BucketClientOptions
```

Inheritance

[object](#)  ← [AWSS3BucketClientOptions](#)

Implements

[IAWSS3BucketClientOptions](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Properties

AccessKey

Gets or sets the AWS key - Can be used to override keys provided by the environment. If deploying inside an EC2 instance AWS keys will already be available via environment variables and don't need to be specified. Follow AWS best security practices on

<https://docs.aws.amazon.com/general/latest/gr/aws-access-keys-best-practices.html> .

```
public string? AccessKey { get; set; }
```

Property Value

[string](#) 

AccessSecret

Gets or sets the AWS endpoint - used to override the default service endpoint. If deploying inside an EC2 instance AWS keys will already be available via environment variables and don't need to be specified. Follow AWS best security practices on

<https://docs.aws.amazon.com/general/latest/gr/aws-access-keys-best-practices.html>.

```
public string? AccessSecret { get; set; }
```

Property Value

string

BucketName

Gets or sets the AWS bucket name. Cannot be `null` when `S3ClientFactory` is not set.

```
public string BucketName { get; set; }
```

Property Value

string

Endpoint

Gets or sets the AWS endpoint - used for testing to over region endpoint allowing it to be set to localhost.

```
public string? Endpoint { get; set; }
```

Property Value

string

Region

Gets or sets the AWS region endpoint (us-east-1/us-west-1/ap-southeast-2).

```
public string? Region { get; set; }
```

Property Value

string[↗]

S3ClientFactory

Gets or sets a custom factory method to create an [AmazonS3BucketClient](#).

```
public Func<IAWSS3BucketClientOptions, IServiceProvider, AmazonS3BucketClient>?  
S3ClientFactory { get; set; }
```

Property Value

Func[↗]<IAWSS3BucketClientOptions, IServiceProvider[↗], AmazonS3BucketClient>

Timeout

Gets or sets a value indicating the timeout for the S3 client. If the value is set, the value is assigned to the Timeout property of the HttpRequest/HttpClient object used to send requests.

```
public TimeSpan? Timeout { get; set; }
```

Property Value

TimeSpan[↗]?

UseAccelerateEndpoint

Gets or sets a value indicating whether the S3 accelerate endpoint is used. The feature must be enabled on the bucket. Follow AWS instruction on

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/transfer-acceleration.html>.

```
public bool UseAccelerateEndpoint { get; set; }
```

Property Value

[bool](#)

Class AWSS3StorageImageProvider

Namespace: [SixLabors.ImageSharp.Web.AWS.Providers](#)

Assembly: [SixLabors.ImageSharp.Web.Providers.AWS.dll](#)

Returns images stored in AWS S3.

```
public class AWSS3StorageImageProvider : IImageProvider, IDisposable
```

Inheritance

[object](#)  ← [AWSS3StorageImageProvider](#)

Implements

[IImageProvider](#), [IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

AWSS3StorageImageProvider(IOptions<AWSS3StorageImageProviderOptions>, FormatUtilities, IServiceProvider)

Initializes a new instance of the [AWSS3StorageImageProvider](#) class.

```
public AWSS3StorageImageProvider(IOptions<AWSS3StorageImageProviderOptions> storageOptions,  
    FormatUtilities formatUtilities, IServiceProvider serviceProvider)
```

Parameters

storageOptions [IOptions](#)  <[AWSS3StorageImageProviderOptions](#)>

The S3 storage options

formatUtilities [FormatUtilities](#)

Contains various format helper methods based on the current configuration.

`serviceProvider` [IServiceProvider](#)

The current service provider.

Properties

Match

Gets or sets the method used by the resolver to identify itself as the correct resolver to use.

```
public Func<HttpContext, bool> Match { get; set; }
```

Property Value

[Func](#)<[HttpContext](#), [bool](#)>

ProcessingBehavior

Gets the processing behavior.

```
public ProcessingBehavior ProcessingBehavior { get; }
```

Property Value

[ProcessingBehavior](#)

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

Dispose(bool)

Releases the unmanaged resources used by the `AWSS3StorageImageProvider` and optionally releases the managed resources.

```
protected virtual void Dispose(bool disposing)
```

Parameters

`disposing bool` [↗](#)

true to release both managed and unmanaged resources; false to release only unmanaged resources.

GetAsync(HttpContext)

Gets the image resolver associated with the context.

```
public Task<IImageResolver?> GetAsync(HttpContext context)
```

Parameters

`context HttpContext` [↗](#)

The current HTTP request context.

Returns

`Task` [↗](#) `<IImageResolver>`

The `IImageResolver`.

IsValidRequest(HttpContext)

Gets a value indicating whether the current request passes sanitizing rules.

```
public bool IsValidRequest(HttpContext context)
```

Parameters

context [HttpContext](#) 

The current HTTP request context.

Returns

[bool](#) 

The [bool](#) 

Class AWSS3StorageImageProviderOptions

Namespace: [SixLabors.ImageSharp.Web.AWS.Providers](#)

Assembly: [SixLabors.ImageSharp.Web.Providers.AWS.dll](#)

Configuration options for the [AWSS3StorageImageProviderOptions](#) provider.

```
public class AWSS3StorageImageProviderOptions
```

Inheritance

[object](#)  ← [AWSS3StorageImageProviderOptions](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Properties

S3Buckets

Gets or sets the collection of blob container client options.

```
public ICollection<AWSS3BucketClientOptions> S3Buckets { get; set; }
```

Property Value

[ICollection](#)  <[AWSS3BucketClientOptions](#)>

Namespace SixLabors.ImageSharp.Web. AWS.Resolvers

Classes

AWSS3StorageCacheResolver

Provides means to manage image buffers within the [AWSS3StorageCacheResolver](#).

AWSS3StorageImageResolver

Provides means to manage image buffers within the AWS S3 file system.

Class AWSS3StorageCacheResolver

Namespace: [SixLabors.ImageSharp.Web.AWS.Resolvers](#)

Assembly: [SixLabors.ImageSharp.Web.Providers.AWS.dll](#)

Provides means to manage image buffers within the [AWSS3StorageCacheResolver](#).

```
public class AWSS3StorageCacheResolver : IImageCacheResolver
```

Inheritance

[object](#)  ← [AWSS3StorageCacheResolver](#)

Implements

[IImageCacheResolver](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

AWSS3StorageCacheResolver(IAmazonS3, string, string, MetadataCollection)

Initializes a new instance of the [AWSS3StorageCacheResolver](#) class.

```
public AWSS3StorageCacheResolver(IAmazonS3 amazonS3, string bucketName, string imagePath, MetadataCollection metadata)
```

Parameters

[amazonS3](#) [IAmazonS3](#)

The Amazon S3 Client

[bucketName](#) [string](#) 

The bucket name.

`imagePath` `string` 

The image path.

`metadata` `MetadataCollection`

The metadata collection.

Methods

GetMetadataAsync()

Asynchronously gets metadata associated with this image.

```
public Task<ImageCacheMetadata> GetMetadataAsync()
```

Returns

`Task`  `<ImageCacheMetadata>`

The `ImageCacheMetadata`.

OpenReadAsync()

Asynchronously gets the input image stream.

```
public Task<Stream> OpenReadAsync()
```

Returns

`Task`  `<Stream`  `>`

The `Task<TResult>` .

Class AWSS3StorageImageResolver

Namespace: [SixLabors.ImageSharp.Web.AWS.Resolvers](#)

Assembly: [SixLabors.ImageSharp.Web.Providers.AWS.dll](#)

Provides means to manage image buffers within the AWS S3 file system.

```
public class AWSS3StorageImageResolver : IImageResolver
```

Inheritance

[object](#)  ← [AWSS3StorageImageResolver](#)

Implements

[IImageResolver](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

AWSS3StorageImageResolver(IAmazonS3, string, string, GetObjectMetadataResponse?)

Initializes a new instance of the [AWSS3StorageImageResolver](#) class.

```
public AWSS3StorageImageResolver(IAmazonS3 amazonS3, string bucketName, string imagePath,  
    GetObjectMetadataResponse? metadataResponse = null)
```

Parameters

[amazonS3](#) [IAmazonS3](#)

The Amazon S3 Client

[bucketName](#) [string](#) 

The bucket name.

`imagePath` `string` 

The image path.

`metadataResponse` `GetObjectMetadataResponse`

Optional metadata response.

Methods

GetMetadataAsync()

Asynchronously gets metadata associated with this image.

```
public Task<ImageMetadata> GetMetadataAsync()
```

Returns

`Task`  `<ImageMetadata>`

The `ImageMetadata`.

OpenReadAsync()

Asynchronously gets the input image stream.

```
public Task<Stream> OpenReadAsync()
```

Returns

`Task`  `<Stream`  `>`

The `Task<TResult>` .